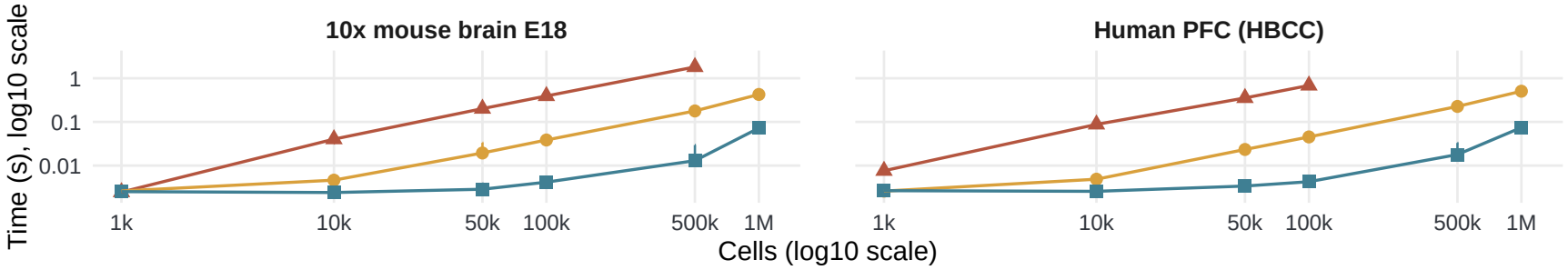
Expression backend scaling

A

Interactive single−gene latency

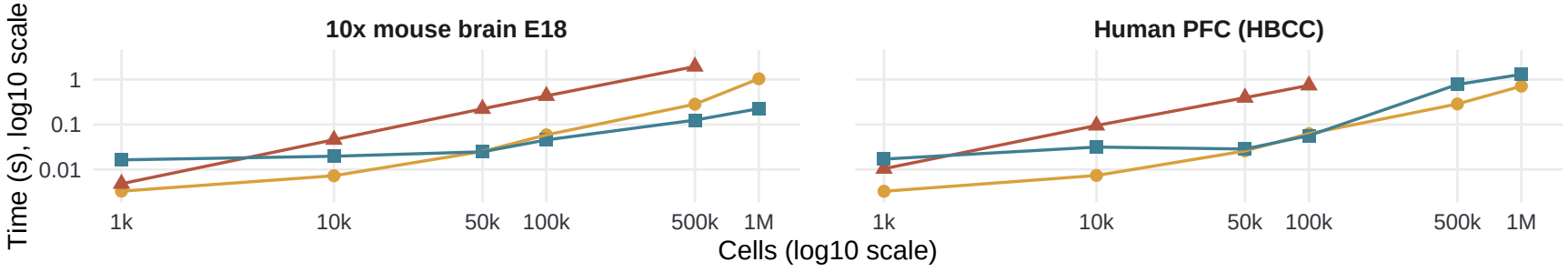
warmed expression lookup; median and range; log scale



B

Marker−panel block latency

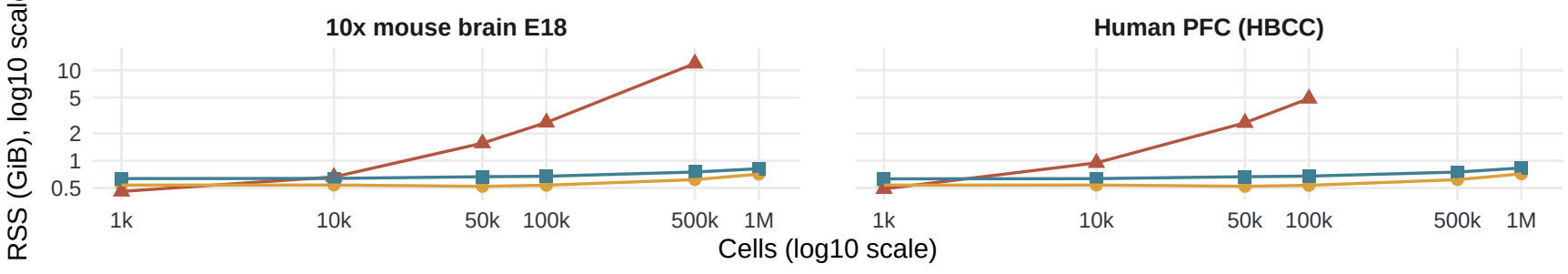
12 genes across all cells; median and range; log scale



C

Resident memory after load and attach

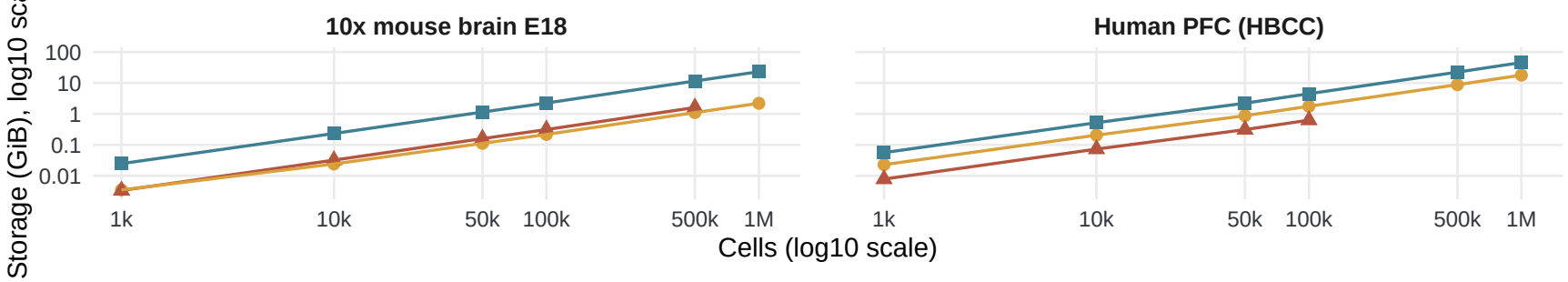
loaded process memory, not build−time peak; median and range



D

Total on−disk footprint

CRB plus external expression data; median and range



Both axes: log10. Points: medians. Whiskers: min−max, not confidence intervals. Lower is better.
Per completed source/tier/backend: 5 builds; 10 read processes (nested within builds).
Embedded skipped: Human PFC (HBCC) 500k: estimated memory exceeds safe budget.
Embedded at 1M was not scheduled in either dataset. Lines join measured tiers; gaps are not measurements.